

Subject Description Form

Subject Code	AAE5204
Subject Title	Autonomous Flight - Mechanics and Control
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with the key knowledge relevant to the flight mechanics, dynamics, and control. 2. To provide students with the capacity to formulate the flight control system by using modern engineering tools and algorithms. 3. To provide students with the knowledge about intelligent planning and control methods to achieve autonomous flight for manned or unmanned aircraft.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand fundamental concepts aircraft coordinate systems and forces; b. able to analysis the longitudinal and lateral direction flight mechanics; c. evaluate aircraft flight stability, controllability and handling quality; d. understand classic and modern flight control system; e. understand search-based and sample-based planning methods and trajectory generation methods; and f. extend their knowledge to analyse and develop new modules or algorithms for desired autonomous flight by flight simulation.
Subject Synopsis/ Indicative Syllabus	Aircraft Six Degrees of Freedom (6-DOF) Equations of Motion: Aircraft coordinate systems; Kinematic model; Dynamic model; Propulsion system model; Model linearization method Longitudinal and Lateral Flight Dynamics and Control: Longitudinal motion and mode approximations; Lateral motion and mode approximations; Handling quality Classic and Modern Flight Control System: Classic flight control system; Modern flight control system; State space modelling; Stability, controllability and observability; State feedback design and optimal control Planning for Autonomous Flight: Global path planning methods including search-based methods and sample-based methods; Local smooth trajectory generation methods Autopilot System Integration and Flight Simulation: Open-source flight controller; Flight simulation platform; Programming and hardware

	interface; Implementation of control and planning algorithms; Introduction to autonomous aerial robotic system																																														
Teaching/Learning Methodology	The teaching and learning methods include lectures, assignment, test, mini project and examination. The tutorials and case study are aimed at providing students with integrated knowledge required for unmanned aircraft systems. Technical/practical examples and problems will be raised and discussed in class/hands on sessions. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="6">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial/Case Study</td><td>√</td><td></td><td>√</td><td></td><td>√</td><td></td></tr></table>	Teaching/Learning Methodology	Outcomes						a	b	c	d	e	f	Lecture	√	√	√	√	√	√	Tutorial/Case Study	√		√		√																				
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Assignment/Test</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Mini Project</td><td>30%</td><td>√</td><td>√</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Examination</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × Continuous Assessment + 0.5 × Final Examination Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by continuous assessment including assignment, closed-book test and mini-project. The continuous assessment is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus. Mini-project is used to assess the students’ capacities of self-learning and problem-solving and effective communication skill in English so as to fulfil the requirements of being aircraft design engineers.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Assignment/Test	20%	√	√	√	√	√		2. Mini Project	30%	√	√		√	√	√	3. Examination	50%	√	√	√	√	√	√	Total	100%						
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Student Study Effort Expected	Class contact:	
	▪ Lecture	27 Hrs.
	▪ Tutorial/case study	12 Hrs.
	Other student study effort:	
	▪ Course work and mini project	42 Hrs.
	▪ Self-study	25 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	1. Pamadi B.N. Performance, stability, dynamics, and control of airplanes. AIAA, 2015. 2. Stevens B.L., Lewis F.L., Johnson E.N., Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Wiley, 2015. 3. Nonami K., Kendoul F., Suzuki S., Wang W., Nakazawa D., Autonomous flying robots: unmanned aerial vehicles and micro aerial vehicles, Springer, 2010.	

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